

SEC P vs NP — Frameworks (live)

SEC (autonomous) — attributed to Ludovico Kubler

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Summary table

Framework ID	Status	Fitness	Generation	Parent	Name
fw_b9e7d103d0	ELABORATING	0.000	0	-	Ergodic Circuit Frame- work
fw_28b4bfb95f	ELABORATING	0.000	0	-	TROPICAL_FOURIER
fw_85a254b4a0	ELABORATING	0.000	0	-	Coarse Geomet- ric Karchmer- Wigderson (CG-KW)
fw_6997a27304	ELABORATING	0.000	0	-	Coarse Geomet- ric Lifting (CGL)
fw_a1a152ae17	ELABORATING	0.000	0	-	Tropical Circuit Weight Analysis (TCWA)
fw_6ef88ddbf3	ELABORATING	0.000	0	-	Coarse- Geometric KW Bounds (CG-KW)

Details

Ergodic Circuit Framework (fw_b9e7d103d0)

- **Status:** ELABORATING
- **Fitness:** 0.000

- **Taxonomy:** COMM_COMPLEXITY
- **Target invariant:** communication_entropy_barrier (real number ≥ 0) $\rightarrow$ bounds The minimum communication complexity in a k-party number-on-forehead model, where lower bounds are derived from the growth rate of Kolmogorov-Sinai entropy in associated dynamical circuits. Aims to prove $\omega(\log n)$ lower bounds for explicit functions like Disjointness, avoiding natural proofs via non-uniform dynamics.

Primitives: - measurable_dynamical_circuit (tuple (C, X, μ, T)): A Boolean circuit C where gates are labeled by operations over a probability space (X, μ) , and $T: X \rightarrow X$ is a measure-preserving transformation. - orbit_signature (function: $N \rightarrow [0,1]$): For a fixed input $x \in \{0,1\}^n$, the orbit signature is the sequence $(\mu(C(T^k(x))))_k$, capturing how the circuit's output evolves under repeated application. - cross_correlation_flow (matrix $\in R^{d \times d}$): For a depth- d circuit, this matrix $F_{i,j} = \int |C_i(x) - C_j(T^j(x))|^2 d\mu(x)$ measures how perturbations via T propagate through layers. Computable via

Tentative axioms: - A1: Any circuit with $o(n)$ communication complexity induces a dynamical system with sublinear Kolmogorov-Sinai entropy growth. - A2: Measure-preserving transformations corresponding to AC^0 circuits have bounded mixing time, while those for PSPACE-complete computations exhibit super-polynomial mixer-profile decay. - A3: If a family of circuits computes a function with high cross_correlation_flow norm, then its communication complexity is $\Omega(n^\delta)$ for some $\delta > 0$.

TROPICAL_FOURIER_ANALYSIS (fw_28b4bfb95f)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** FOURIER_ANALYTIC
- **Target invariant:** MinimalFourierCoefficient (RealNumber) $\rightarrow$ bounds The discrepancy of a function under tropical Fourier analysis.

Primitives: - TropicalPolynomial (Function): A function defined over a tropical semiring (max-plus or min-plus) with computable coefficients. - TropicalFourierTransform (Transformation): A mapping from tropical polynomials to their Fourier coefficients over a tropical semiring. - DiscrepancyMeasure (Metric): A function quantifying the deviation of a tropical polynomial from uniformity.

Tentative axioms: - A1: TropicalConvolution preserves the tropical semiring structure under composition. - A2: TropicalFourierTransform is invertible when restricted to certain tropical polynomials. - A3: DiscrepancyMeasure is bounded by the maximum absolute value of Fourier coefficients.

Coarse Geometric Karchmer-Wigderson (CG-KW) (fw_85a254b4a0)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** KARCHMER_WIGDERSON
- **Target invariant:** Coarse depth $\kappa(f)$ (non-negative real) $\rightarrow$ bounds $\kappa(f) := \sup$ over nontrivial $c \in HX^1(X_f)$ and $T \in C_u^*[X_f]$ of $\log|\langle c, T \rangle| / \log(\text{propagation}(T))$. The conjecture is $\text{depth}(f) \geq \kappa(f)$, so a super-logarithmic $\kappa(f)$ for an explicit f gives a super-logarithmic formula-depth lower bound (a $P \not\subseteq NC^1$ separation if κ is poly-large).

Primitives: - KW-metric space (finite metric space (X_f, d_f) with $X_f = f^{-1}\{0\} \sqcup f^{-1}\{1\}$): For a boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$, define $d_f(x,y) = \log_2(\text{min number of coordinates a depth-bounded distinguisher must read to certify } f(x) \neq f(y))$. - Controlled cover (finite family $U = \{U_i \subset X_f\}$ with diameter bound R): A cover whose parts have d_f -diameter at most R and whose overlap multiplicity is at most m . Encoded as a hypergraph H_U (vertices = X_f , edges = U_i). - Coarse cocycle (function $c: X_f \times X_f \rightarrow \mathbb{Z}$ with finite support modulo controlled equivalence): A discrete 1-cochain in the Roe-style coarse cohomology $HX^1(X_f)$ restricted to the KW pair structure. Stored

as a sparse table on pairs (x,y) with d_f - Asymptotic-dimension witness (vertex coloring $\chi : X_f \rightarrow [k]$ together with diameter bound R): Witnesses $\text{asdim}(X_f) \leq k-1$ in the sense of Gromov: each color class decomposes into d_f -bounded pieces of diameter $\leq R$ that are pairwise R -disjoint. C - Roe-controlled operator (matrix $T \in R^{\{X_f \times X_f\}}$ with $T_{\{x,y\}} = 0$ whenever $d_f(x,y) > R$): Element of the algebraic uniform Roe algebra $C_u^*[X_f]$. Stored as a banded sparse matrix in the d_f -metric; multiplication, adjoint, and trace are al

Tentative axioms: - A1 (Coarse-KW link): formula depth $d(f) \geq \Omega(\log(\text{asdim}(X_f)))$ and more strongly $d(f) \geq \Omega(\kappa(f))$; proved by translating a KW protocol into a controlled cover whose multiplicity exponent equals depth. - A2 (Composition sub-additivity): $\text{asdim}(X_{\{f \circ g\}}) \geq \text{asdim}(X_f) + \text{asdim}(X_g) - O(1)$, giving a KRW-style direct-sum theorem at the level of coarse dimension. - A3 (Anti-natural-proofs): for a uniformly random f , $HX^1(X_f)$ is generically zero (Roe-algebraic concentration of measure), so κ is NOT a ‘largeness’ property in the Razborov-Rudich sense — random f ’s - A4 (Anti-relativization): κ is a metric invariant of d_f and is destroyed by oracle access (oracle gates collapse d_f to ≤ 1), so κ -based bounds cannot relativize, in the spirit of Mendel-Naor metric - A5 (Roe-index rigidity): nontrivial classes in $HX^1(X_f)$ detected by the Roe-trace pairing are stable under coarse equivalence, mirroring Yu’s coarse Baum-Connes results (Yu 2000; Nowak-Yu 2012); expl

Coarse Geometric Lifting (CGL) (fw_6997a27304)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** LIFTING
- **Target invariant:** CoarseLiftingComplexity (CLC) (N -valued function of $(f, G) \rightarrow$ bounds The deterministic 2-party communication complexity of $f \circ G^n$. Specifically: $CC(f \circ G^n) \geq Q(f) \cdot \log_2(\text{asdim}_R(G)+1)$ for the gadget scale $R = \text{diam}(G)$, with the goal of recovering and surpassing known query-to-communication lifting bounds (Raz-McKenzie, Göös-Pitassi-Watson) for gadgets where positive asymptotic dimension can be certified.

Primitives: - MetricGadget ($(X: \text{FiniteSet}, Y: \text{FiniteSet}, g: X \times Y \rightarrow \{0,1\}, d: (X \times Y)^2 \rightarrow \mathbb{R}_{\geq 0})$): A boolean gadget g equipped with a graph/Hamming metric d on $X \times Y$. Implementable as a tuple of two finite sets, a truth-table for g , and a precomputed - LiftedInputSpace $((X \times Y)^n, d_{\oplus})$: The n -fold product of a MetricGadget under the $\square^1$ (Hamming-sum) product metric $d_{\oplus}((u_i), (v_i)) = \sum d(u_i, v_i)$. Implementable as a generator over tup - RoeCover ($\text{List}[\text{Subset}((X \times Y)^n)]$ with multiplicity m and scale R): A finite cover of the lifted input space whose members have $d_{\oplus}$ -diameter $\leq R$ and any point is in $\leq m$ members. This is the discrete witness used in the - ProtocolPullback ($\text{Dict}[\text{transcript} \rightarrow \text{Subset}((X \times Y)^n)]$): The partition of inputs induced by transcripts of a (deterministic or randomized) communication protocol Π , viewed as a candidate RoeCover. Implementa - FolnerWitness ($\text{Sequence}[\text{Subset}((X \times Y)^n)]$): A nested family $F_1 \subset F_2 \subset \dots$ with $|\partial_R F_k|/|F_k| \rightarrow 0$ for each scale R , certifying amenability/Property A of the lifted space relative to a gadget.

Tentative axioms: - A1 (Coarse-Lift Functoriality): If two gadgets G_1, G_2 are coarsely equivalent (a bi-Lipschitz bijection up to bounded error), then $\text{CLC}(f, G_1) = \Theta(\text{CLC}(f, G_2))$ for every f . - A2 (Protocol $\rightarrow$ Cover): Every deterministic protocol Π for $f \circ G^n$ of cost c yields a RoeCover with multiplicity $\leq 2^c$ and bounded scale R_{Π} depending only on G ; hence $CC(f \circ G^n) \geq \log_2(\min \text{multiplicity})$ - A3 (Asdim Amplification): $\text{asdim}(G_1 \otimes G_2) \geq \text{asdim}(G_1) + \text{asdim}(G_2) - O(1)$, so iterated tensoring of a single ‘good’ gadget produces unbounded coarse dimension. - A4 (Property-A Transfer): If G has Property A and f has query complexity $Q(f)$, then $\text{CLC}(f, G) \geq Q(f) \cdot h(G)$ where $h(G)$ is the Hilbert-space compression exponent of G — a non-natural, non-relativizing - A5 (Non-Algebrization): The Roe algebra $C^*_R((X \times Y)^n)$ is not closed under polynomial extensions of the gadget alphabet; therefore lower bounds derived from its K -theory do not algebrize.

Tropical Circuit Weight Analysis (TCWA) (fw_a1a152ae17)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** BOUNDED_ARITHMETIC
- **Target invariant:** Tropical Proof Rank (integer) $\rightarrow$ bounds The minimum number of phase transitions required to simulate a bounded arithmetic proof line in V^0 or IA_0 , thereby bounding the strength of definable functions and aiming to separate bounded arithmetic theories via circuit phase complexity.

Primitives: - Tropical Circuit (tuple (G, w)): A directed acyclic graph G with nodes labeled as inputs (variables or constants), tropical addition (min or max), or tropical multiplication (usual ad
- Tropical Derivation (function $\delta: \text{Circuit} \rightarrow \text{Circuit}$): A formal derivative operator on tropical circuits: for a node computing $\min(x, y)$, the derivation selects the active operand (argmin) with tie-breaking - Weight Profile (list of integers): For a tropical circuit C and input $x \in \mathbb{Z}^n$, the weight profile is the sequence of edge weights traversed during evaluation under tropical semantics, or
- Circuit Phase Space (subset of $\mathbb{R}^n$): Partition of input space $\mathbb{R}^n$ into convex polyhedral regions where the tropical circuit's active computation path (i.e., argmin/argmax decisions) remain

Tentative axioms: - A1: (Phase Stability Axiom) Any function provably total in IA_0 has a tropical circuit with polynomially many phase cells under uniform input scaling. - A2: (Weight-Proof Correspondence) The sum of absolute weights in a tropical circuit for a formula encoding a proof line is at least the bit-complexity of the cut-free proof in bounded arithmetic. - A3: (Tropical Soundness) If a bounded arithmetic theory proves a $\forall \Sigma_1^b$ sentence, then its tropical circuit model exhibits a homotopy-stable phase space under perturbation.

Coarse-Geometric KW Bounds (CG-KW) (fw_6ef88ddb3)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** KARCHMER_WIGDERSON
- **Target invariant:** formula_depth_D(f) (natural_number) $\rightarrow$ bounds $\log_2$ of minimum size, equivalently minimum depth, of a De Morgan formula computing f . The framework conjectures $D(f) \geq c * \text{asdim}(X_f)$ and $D(f) \geq c' / \alpha * (X_f)$, giving super-logarithmic depth lower bounds when X_f has no Hilbert compression (Yu 2000; Nowak-Yu 2012).

Primitives: - KW_metric_space (metric_space): For boolean $f: \{0,1\}^n \rightarrow \{0,1\}$, the set $X_f = f^{-1}(0) \times f^{-1}(1)$ equipped with the disagreement-Hamming metric $d((x,y), (x',y')) = |\{i : (x_i \neq y_i) \vee (x'_i \neq y'_i)\}|$
- AsymDim_cover (indexed_cover): A family $\{U_a\}_{a \in A}$ of subsets of X_f with uniformly bounded diameter and bounded r -multiplicity (every r -ball meets $\leq k+1$ sets). Encoded as a hy
- Hilbert_compression_embedding (Lipschitz_map): A map $\phi: X_f \rightarrow \mathbb{R}^N$ with explicit upper Lipschitz constant L and lower compression $\rho(t) \sim t^\alpha$, certified by a Gram matrix and PSD check. Compu - Coarse_partition_tree (rooted_labeled_tree): A binary tree whose leaves partition X_f into 'monochromatic' coordinate-rectangles; each internal node is labeled with a coordinate i in $[n]$ separati - Property_A_witness (function_family): Maps $\xi: X_f \rightarrow \ell_1(X_f)$ of finite support such that $\|\xi_x - \xi_y\|_1$ is small when $d(x,y)$ bounded, and supports lie in fixed-radius balls. Verifia

Tentative axioms: - A1 (Tree-to-Dimension): Every depth- d KW protocol yields an AsymDim_cover of X_f with multiplicity $\leq d+1$ and bounded diameter; hence $\text{asdim}(X_f) \leq D(f)$. - A2 (Compression Lower Bound): If X_f admits a Hilbert embedding with compression exponent α , then $D(f) \geq \Omega(1/\alpha)$; contrapositively, expander-like X_f forces large depth (cf. Gromov's a-T-m - A3 (Coarse Composition / KRW): $\text{asdim}(X_{f \circ g}) \geq \text{asdim}(X_f) + \text{asdim}(X_g) - O(1)$, giving a coarse-geometric route to the Karchmer-Raz-Wigderson conjecture. - A4 (Non-Naturalness): The predicate ' $\text{asdim}(X_f) \geq k$ ' is neither large nor constructive in the Razborov-Rudich sense - random functions have bounded asdim with high probability because random metric s - A5 (Non-Relativizing / Non-Algebrizing): Coarse invariants depend on the fine combinatorial structure of $f^{-1}(0)$ and $f^{-1}(1)$ as embedded in the Hamming cube, not on oracle access or low-degree alg